Practice Sheet: UNIQUE Function

This practice sheet helps you master the **UNIQUE** function in Excel.

The UNIQUE function is used to extract unique values from a list. It removes repeated entries and shows each item only once.

In real office work, UNIQUE is useful for preparing lists such as unique customer names, unique cities, unique product categories, unique payment modes, and unique combinations of records.

New Function: UNIQUE

`=UNIQUE(array, [by_col], [exactly_once])`

Meaning:

- **array**: The range from which unique values will be extracted.
- **by_col**: Usually left blank. It is used when comparing columns instead of rows.
- **exactly_once**: Optional. If TRUE, it shows only values that appear exactly once.

Example:

`=UNIQUE(C2:C51)`

This formula extracts the unique values from the range **C2:C51**.

Important Note: The UNIQUE function is available in newer versions of Excel such as **Excel 2021** and Microsoft 365.

Step 1: Set Up the Main Data Table

In **Sheet1**, starting from cell **A1**, enter the following table.

Order ID	Customer Name	City	Category	Payment Mode	Qty	Amount (₹)
O001	Ankit Sharma	Guwahati	Electronics	UPI	2	18500
O002	Rima Das	Jorhat	Stationery	Cash	5	1250
O003	Tanmay Bora	Dibrugarh	Grocery	Card	8	3400
O004	Priyanka Dutta	Silchar	Clothing	UPI	3	2700
O005	Arindam Saikia	Guwahati	Electronics	Card	1	42000
O006	Moumita Roy	Tezpur	Grocery	Cash	6	2100
O007	Bijoy Kalita	Nagaon	Furniture	UPI	1	9500
O008	Nisha Sharma	Guwahati	Stationery	UPI	10	1600
O009	Imran Ali	Jorhat	Clothing	Cash	2	1800
O010	Rakesh Nath	Dibrugarh	Electronics	Card	1	32000
O011	Puja Das	Silchar	Grocery	UPI	7	2800
O012	Akash Baruah	Tezpur	Furniture	Cash	2	14500

Order ID	Customer Name	City	Category	Payment Mode	Qty	Amount (₹)
O013	Manisha Paul	Guwahati	Clothing	Card	4	3600
O014	Rohit Singh	Nagaon	Electronics	UPI	1	25500
O015	Sangeeta Gogoi	Jorhat	Stationery	Cash	12	2400
O016	Debajit Deka	Guwahati	Grocery	Card	9	3900
O017	Kavita Roy	Silchar	Furniture	UPI	1	12000
O018	Suman Chetia	Dibrugarh	Clothing	Cash	3	3300
O019	Farhana Begum	Tezpur	Electronics	UPI	2	47000
O020	Kunal Sharma	Nagaon	Grocery	Card	5	2200
O021	Deepika Kalita	Guwahati	Stationery	Cash	15	3000
O022	Rajib Das	Jorhat	Furniture	Card	1	18000
O023	Neha Kapoor	Silchar	Electronics	UPI	1	29500
O024	Hiren Bora	Dibrugarh	Grocery	Cash	4	1750
O025	Alisha Ahmed	Tezpur	Clothing	Card	5	4100
O026	Pranjal Saikia	Guwahati	Furniture	UPI	2	22500
O027	Karishma Dey	Nagaon	Stationery	Cash	6	1350
O028	Vikram Sinha	Jorhat	Electronics	Card	1	52000
O029	Meera Nair	Silchar	Grocery	UPI	8	3600
O030	Abhishek Roy	Dibrugarh	Clothing	Cash	2	2600
O031	Sneha Dutta	Tezpur	Stationery	UPI	9	1900
O032	Rahul Choudhury	Guwahati	Electronics	Card	2	61500
O033	Ankita Mahanta	Nagaon	Grocery	Cash	10	4200
O034	Nilotpal Das	Jorhat	Furniture	UPI	1	15500
O035	Jyoti Devi	Silchar	Clothing	Card	4	3800
O036	Manash Sarma	Dibrugarh	Electronics	UPI	1	35000
O037	Rituparna Bora	Tezpur	Grocery	Cash	6	2400
O038	Wasim Ahmed	Guwahati	Stationery	Card	11	2750
O039	Pankaj Dutta	Nagaon	Electronics	UPI	1	28500
O040	Sonali Gogoi	Jorhat	Clothing	Cash	3	3100
O041	Chandan Roy	Silchar	Furniture	Card	2	24000

Order ID	Customer Name	City	Category	Payment Mode	Qty	Amount (₹)
O042	Bipasha Das	Dibrugarh	Grocery	UPI	7	2950
O043	Nirmal Kalita	Tezpur	Electronics	Card	1	44500
O044	Aditi Sharma	Guwahati	Clothing	Cash	5	4600
O045	Rajashree Deka	Nagaon	Stationery	UPI	8	1650
O046	Partha Pratim	Jorhat	Furniture	Card	1	21000
O047	Mitali Saikia	Silchar	Grocery	Cash	9	3750
O048	Saurav Bora	Dibrugarh	Electronics	UPI	2	58000
O049	Monika Paul	Tezpur	Clothing	Card	2	2200
O050	Diganta Gogoi	Guwahati	Grocery	UPI	6	2600

Step 2: Create Sheet2 for Unique Results

Create a new sheet named **Unique Results**.

All UNIQUE formulas should be entered in the **Unique Results** sheet. This keeps the original sales data safe in Sheet1.

Step 3: Extract Unique Cities

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A1	Unique Cities
A2	`=UNIQUE(Sheet1!C2:C51)`

This extracts each city name only once.

Step 4: Extract Unique Product Categories

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
D1	Unique Categories
D2	`=UNIQUE(Sheet1!D2:D51)`

This extracts each product category only once.

Step 5: Extract Unique Payment Modes

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
G1	Unique Payment Modes
G2	`=UNIQUE(Sheet1!E2:E51)`

This extracts each payment mode only once.

Step 6: Extract Unique Customer Names

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
J1	Unique Customer Names
J2	`=UNIQUE(Sheet1!B2:B51)`

This extracts each customer name only once.

Step 7: Sort Unique Cities

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A60	Sorted Unique Cities
A61	`=SORT(UNIQUE(Sheet1!C2:C51))`

This first extracts the unique cities and then sorts them alphabetically.

Step 8: Sort Unique Categories

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
D60	Sorted Unique Categories
D61	`=SORT(UNIQUE(Sheet1!D2:D51))`

This first extracts the unique categories and then sorts them alphabetically.

Step 9: Sort Unique Payment Modes

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
G60	Sorted Unique Payment Modes
G61	`=SORT(UNIQUE(Sheet1!E2:E51))`

This first extracts the unique payment modes and then sorts them alphabetically.

Step 10: Extract Unique City and Category Combinations

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A90	Unique City and Category Combinations
A91	`=UNIQUE(Sheet1!C2:D51)`

This extracts unique combinations of **City** and **Category**.

Step 11: Extract Unique Category and Payment Mode Combinations

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
E90	Unique Category and Payment Mode Combinations
E91	`=UNIQUE(Sheet1!D2:E51)`

This extracts unique combinations of **Category** and **Payment Mode**.

Step 12: Extract Unique Cities from Electronics Orders

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A155	Unique Cities for Electronics
A156	`=UNIQUE(FILTER( Sheet1!C2:C51,Sheet1!D2:D51="Electronics", "No Records Found"))`

This first filters only Electronics orders and then extracts the unique cities from them.

Step 13: Extract Unique Customers Who Paid by UPI

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
D155	Unique UPI Customers
D156	`=UNIQUE(FILTER( Sheet1!B2:B51,Sheet1!E2:E51="UPI", "No Records Found"))`

This first filters only UPI orders and then extracts the unique customer names.

Step 14: Extract Unique Categories from Guwahati Orders

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
G155	Unique Categories in Guwahati
G156	`=UNIQUE(FILTER( Sheet1!D2:D51,Sheet1!C2:C51="Guwahati", "No Records Found"))`

This first filters only Guwahati orders and then extracts the unique product categories.

Step 15: Sort Unique Customers Who Bought Electronics

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A200	Sorted Unique Electronics Customers
A201	`=SORT(UNIQUE(FILTER( Sheet1!B2:B51,Sheet1!D2:D51="Electronics", "No Records Found")))`

This extracts unique Electronics customers and sorts them alphabetically.

Step 16: Sort Unique Cities with Amount Above ₹20,000

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
D200	Sorted Unique High Value Cities
D201	`=SORT(UNIQUE(FILTER( Sheet1!C2:C51,Sheet1!G2:G51>20000, "No Records Found")))`

This extracts unique cities where the order amount is above **₹20,000**.

Step 17: Extract Unique Categories Bought by Cash Customers

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
G200	Unique Cash Categories
G201	`=UNIQUE(FILTER( Sheet1!D2:D51,Sheet1!E2:E51="Cash", "No Records Found"))`

This extracts unique product categories from orders paid by Cash.

Step 18: Extract Customer Names That Appear Exactly Once

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
A245	Customers Appearing Exactly Once
A246	`=UNIQUE(Sheet1!B2:B51,,TRUE)`

This shows only those customer names that appear exactly once in the list.

Step 19: Unique Customers with Quantity More Than 5

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
D245	Unique Customers Qty Above 5

D246	`=UNIQUE(FILTER(`
	 `Sheet1!B2:B51,Sheet1!F2:F51>5,`
	 `"No Records Found"))`

This extracts unique customers whose order quantity is more than **5**.

Step 20: Unique Cities for UPI Electronics Orders

In the **Unique Results** sheet, enter the following:

Cell	Value to Enter
G245	Unique Cities for UPI Electronics

G246	`=UNIQUE(FILTER(`
	 `Sheet1!C2:C51,`
	 `(Sheet1!D2:D51="Electronics")*(Sheet1!E2:E51="UPI"),`
	 `"No Records Found"))`

This first filters orders where category is **Electronics** and payment mode is **UPI**, then extracts the unique cities.

Practice Questions

Use the **UNIQUE** function to solve the following questions. Use the exact cells given below in the **Unique Results** sheet so that everyone's workbook remains uniform.

Cell	Task
A300	Extract unique **Customer Names**.
A360	Extract unique **Cities**.
A420	Extract unique **Categories**.

Cell	**Task**
A480	Extract unique **Payment Modes**.
A540	Extract unique combinations of **City and Payment Mode**.
A600	Extract unique combinations of **Customer Name and Category**.
A660	Extract sorted unique **Customer Names**.
A720	Extract sorted unique **Cities**.
A780	Extract sorted unique **Categories**.
A840	Extract unique customers from **Electronics** orders.
A900	Extract unique customers from **Grocery** orders.
A960	Extract unique customers from **Clothing** orders.
A1020	Extract unique cities from orders paid by **UPI**.
A1080	Extract unique cities from orders paid by **Cash**.
A1140	Extract unique cities from orders paid by **Card**.
A1200	Extract unique categories where order amount is above **₹20,000**.
A1260	Extract unique customer names where order quantity is more than **5**.
A1320	Extract unique cities where category is **Furniture**.
A1380	Extract unique payment modes used in **Guwahati**.
A1440	Extract unique categories purchased in **Jorhat**.
A1500	Extract sorted unique cities for **Electronics** orders.
A1560	Extract sorted unique customers who paid through **UPI**.
A1620	Extract unique city and category combinations for orders above **₹10,000**.
A1680	Extract unique customers who bought **Electronics** and paid through **Card**.
A1740	Extract unique cities where category is **Grocery** and quantity is more than **5**.

Extra Challenge

Create a new sheet named **Dynamic Unique**.

In the **Dynamic Unique** sheet, create the following criteria table in cells **A1:B3**:

Cell	Value to Enter
A1	Criteria
B1	Value
A2	Category
B2	Electronics
A3	Payment Mode
B3	UPI

In cell **A5**, type:

```
Unique Cities Based on Criteria
```

In cell **A6**, enter the following formula:

```
=SORT(UNIQUE(FILTER(
Sheet1!C2:C51,(Sheet1!D2:D51=B2)*(Sheet1!E2:E51=B3),
"No Records Found")))
```

This formula filters orders according to the selected category and payment mode, extracts the unique cities, and sorts them alphabetically.

Now change the values in cells **B2** and **B3** and observe how the result changes automatically.

Homework

Create a new sheet named **Homework**. In the Homework sheet, create a new table of **50 students** starting from cell **A1** with the following columns:

Roll No.	Student Name	Course	Batch	City	Marks	Status

Then use the **UNIQUE**, **SORT**, and **FILTER** functions to find the following. Use the exact cells given below in the **Homework** sheet.

Cell	Task
I1	Extract unique **Courses**.
I60	Extract unique **Batches**.
I120	Extract unique **Cities**.
I180	Extract unique **Status** values.
I240	Extract sorted unique **Courses**.
I300	Extract sorted unique **Cities**.
I360	Extract unique student names from the **DCA** course.
I420	Extract unique cities from the **PGDCA** course.
I480	Extract unique batches from **Active** students.
I540	Extract unique courses where marks are above **80**.

Cell	**Task**
I600	Extract unique cities where marks are below **50**.
I660	Extract unique student names from the **Morning** batch.
I720	Extract unique course and batch combinations.
I780	Extract unique city and status combinations.
I840	Extract sorted unique cities for students who are **Active**.

Save your workbook as **Excel_Practice_Module16.xlsx** and submit it to your instructor.

End of Module 16